

=== CONFIGURACION GNB (1780308529) ===

```
# gNB configuration for URLLC VR E-Sports
# Optimized for low latency and high bandwidth for VR applications

mcc: 001
mnc: 010
n_nci: 0x100000001
tac: 1

amf_addr: 10.53.1.2
amf_port: 38412

# Radio configuration
# Based on RAG context for Band 28 and srate
srate: 23.04e6
nof_prb: 25
bands: [28]
dl_nr_arfcn: 154000
ssb_nr_arfcn: 154090
tx_gain: 70
rx_gain: 40
channel_bandwidth_MHz: 5

# CU_CP configuration
cu_cp:
  bind_addr: 10.53.1.3
  sctp_port: 38412

# ZMQ configuration for communication with UE
zmq:
  tx_port: 2000
  rx_port: 2001
  rx_addr: 0.0.0.0

# Slice configuration for URLLC
slices:
  - sst: 1
    sd: 0x000001 # Default slice for VR E-Sports
    qos_characteristics:
      - qci: 1 # QCI 1 for URLLC (conversational voice, high priority)
        priority_level: 1
        preemption_vulnerability: 1
        preemption_capability: 1
        # MBR and GBR set high for VR traffic, using Mbps unit
        mbr:
          downlink:
            value: 802
            unit: 2 # Mbps
          uplink:
            value: 802
            unit: 2 # Mbps
        gbr:
          downlink:
            value: 802
            unit: 2 # Mbps
          uplink:
            value: 802
            unit: 2 # Mbps

# No cu_up or gtpu block as per strict rule
```

=== CONFIGURACION UE (1780308529) ===

UE configuration for URLLC VR E-Sports
Matches gNB radio parameters and core network settings

```
[rf]
freq_offset = 500
tx_gain = 70
rx_gain = 40
srate = 23.04e6
nof_antennas = 1
device_name = uhd
device_args = sync=internal
```

```
[rat.nr]
bands = 28
nof_carriers = 1
max_nof_prb = 25
nof_prb = 25
dl_nr_arfcn = 154000
ssb_nr_arfcn = 154090
```

```
[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap
```

```
[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1
```

```
[usim]
mode = soft
algo = milenage
opc = 504f20634f6320504f50206363500a4f
k = 6874736969202073796d4b2079650a73
imsi = 0010100000000001
```

```
[rrc]
release = 15
ue_category = 4
```

```
[nas]
apn = srsapn
apn_protocol = ipv4
```

```
[gw]
# ip_addr is explicitly prohibited here
# netns = ue1
# ip_devname = tun_srsue
# ip_netmask = 255.255.255.0
```

```
[gui]
enable = true
```

=== DOCKER COMPOSE (1780308529) ===

```
# Copyright 2023-2024 SRS-RAN Project
#
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.
```

version: '3.8'

services:

5gc:

build:

context: .

dockerfile: Dockerfile.5gc

container_name: open5gs-5gc

privileged: true

command: 5gc -c open5gs-5gc.yml

networks:

ran:

ipv4_address: 10.53.1.2

volumes:

- ./configs/open5gs:/open5gs/config

healthcheck:

test: ["CMD", "pgrep", "open5gs-amfd"]

interval: 10s

timeout: 5s

retries: 5

gnb:

build:

context: .

dockerfile: Dockerfile.gnb

container_name: srsran-gnb

privileged: true

command: srsgnb -c gnb.yml

networks:

ran:

ipv4_address: 10.53.1.3

volumes:

- ./configs/srsran:/srsran/config

healthcheck:

test: ["CMD", "pgrep", "srsgnb"]

interval: 10s

timeout: 5s

retries: 5

networks:

ran:

driver: bridge

ipam:

config:

- subnet: 10.53.1.0/24